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RBI's focus on inflation is timely

ON WEDNESDAY, with the Monetary Policy Committee of the Reserve Bank of India raising the repo rate by 25 basis points, it moved up from 5.25 per cent to 5.5 per cent. A higher repo rate is expected to make borrowing more costly for everyone and thus acts as a drag on economic activity. The decision to make the first hike of the repo rate since February 2023 was not only unanimous — all six members of the MPC voted in favour — but also widely expected. The less obvious but perhaps more consequential change was the shift in the RBI's stance from "neutral" to "calibrated tightening". The last time the RBI assumed this stance was exactly eight years ago.

The RBI's primary mandate, as Governor Sanjay Malhotra has underlined, is to ensure price stability while keeping in view economic growth. Since the last policy review in August, the situation has changed on both these counts, necessitating a change in policy. On economic growth, the Indian economy has surprised everyone by showing resilience despite a worsening global context. The 7.8 per cent growth in Gross Domestic Product during the first quarter (April, May, June) surprised on the upside when the data was released at the end of August. This has led the RBI to bump up the full-year (2026-27) forecast from 6.7 per cent, as pencilled in the August policy, to 7.1 per cent now. But notwithstanding the domestic growth momentum, the situation has become considerably more challenging from the point of view of inflation, thanks to the "sudden re-escalation of the West Asia conflict in September and the consequent hardening and volatility in global crude prices". Making the domestic inflation outlook worse are the deficient Southwest monsoon and El Niño conditions. Retail inflation has been going up steadily through the year, breaching the 4 per cent RBI target level. In fact, the RBI now expects it to touch 6 per cent by the end of December with full-year inflation around 5.2 per cent.

Given that growth is robust even as inflation is trending upwards and beyond the RBI target, raising the repo rate in a bid to dampen overall demand in the economy and prevent inflation becoming more broad-based is timely. What is even more welcome is the shift in the RBI's stance. It suggests that the RBI is squarely focussed on dealing with a possible spike in inflation. Simply put, interest rate cuts are off the table for the time being, and the RBI has signalled the start of a rate hike cycle, given that globally other central banks are expected to raise interest rates in the coming months.

The universe whispers through ghost particles

FOR CENTURIES, astronomy has essentially been the science of observing light. However, some of the universe's most consequential events take place in environments so dense or violent that photons cannot escape. Could the universe be observed through something other than light? In 1930, when Austrian physicist Wolfgang Pauli proposed the existence of an elusive particle to explain a mystery in radioactive decay, he was laying the foundations for a new way of exploring the cosmos. The ghost particle called the neutrino was detected two decades later, and the quest to understand it has since been rewarded with a string of Nobel Prizes. On Tuesday, the Royal Swedish Academy of Sciences, which awards the Nobel Prize in Physics, recognised another pioneer in the field — Belgian-American astrophysicist Francis Halzen. His work has used neutrinos to probe some of the universe's hitherto most inaccessible environments, helping open a new window on the cosmos.

Neutrinos have tiny masses, carry no electric charge and interact so weakly with matter that billions pass through the Earth — and our bodies — without leaving a trace. These particles can escape from the dense interiors of exploding stars and other extreme cosmic environments, carrying information about places that conventional telescopes cannot see. Halzen recognised this potential. His pioneering work led to the IceCube Observatory, an enormous neutrino detector buried deep beneath the Antarctic ice. Rather than building a conventional telescope, scientists effectively turned a cubic kilometre of natural ice at the South Pole into a detector, embedding more than 5,000 optical sensors in it. Almost every neutrino passes straight through the ice. But on the rare occasion when one sheds its shyness, the resulting flash of light can give information on the particle's travel path.

In 2013, IceCube provided evidence for a flux of high-energy neutrinos originating beyond the solar system. Halzen's work demonstrated that sometimes the way to see farther is not to build a more powerful eye, but to learn to detect what the universe is whispering through particles that pass silently through it. The next frontier is multi-messenger astronomy, in which neutrinos can be studied alongside light, gravitational waves and cosmic rays. Together, these cosmic messengers could reveal the workings of black holes, exploding stars and other extreme objects that remain invisible to any single form of observation.

Behind the central bank's pivot, need to safeguard price stability



DHARMAKIRTI JOSHI

THE SHIFT in stance by the RBI's Monetary Policy Committee (MPC), accompanied by a rate hike in its October review, was hardly unexpected. Monetary policy is shaped principally by domestic growth and inflation dynamics, with global developments playing a secondary role. Recent developments on both fronts pointed towards a recalibration of policy. This year was marked by an unusual combination of shocks and uncertainties. Yet, despite the conflict in West Asia and unresolved tariff-related turbulence, India's economy has remained remarkably resilient. Growth has stayed robust, while inflation, particularly core inflation, has thus far been relatively contained. However, the balance of risks has begun to shift. Since the MPC's August review, upside risks to inflation have become more pronounced. Three factors stand out.

First, the conflict in West Asia, marked by repeated cycles of escalation and de-escalation, has kept commodity markets volatile. Recent developments have pushed up crude prices again, with the Indian crude basket, as estimated by the Petroleum Planning and Analysis Cell, returning to its April peak. Reflecting this, we now expect crude prices to average $88-93 per barrel this fiscal, compared with our earlier forecast of $82-87 per barrel. Even if geopolitical tensions ease, crude prices may remain elevated as damaged infrastructure is repaired and depleted inventories rebuilt. In such a scenario, another round of pass-through to domestic petrol and diesel prices appears likely. While higher pump prices could temper demand and ease pressure on oil marketing companies, they would also raise inflation through both direct and indirect channels.

Food inflation represents a second major vulnerability. Food prices are the most volatile component of India's inflation basket, and recent increases reflect both base effects and emerging supply concerns. To be sure, the relationship between monsoons and food inflation is not linear. Not every rainfall-deficient year has led to higher food inflation, and the government's free food programme for nearly 800 million people, alongside healthy rice buffer stocks, provides an important cushion.

Nevertheless, the 2026 monsoon has ended with rainfall 13 per cent below the long-period average and over 42 per cent of districts classified as deficient. The problem extends beyond the aggregate shortfall. The spatial distribution and timing of rainfall have been particularly unfavourable, making this one of the past decade's most disrupted monsoon seasons. The implications extend beyond the current kharif season. Weak monsoons can impair rabi production by reducing soil moisture and reservoir levels. As of October 1, reservoir storage was 12.2 per cent below normal and 20.4 per cent lower than a year earlier. Crisil's Deficient Rainfall Impact Parameter (DRIP) points to broad-based stress, with cotton, tur and bajra among the most vulnerable crops.

Intensifying El Niño conditions, combined with supply disruptions linked to the West Asia conflict, are exerting upward pressure on food prices worldwide. Of note is edible oil inflation, driven by weather-related disruptions in Southeast Asia and rising biofuel demand. According to the World Meteorological Organisation, the probability of a strong El Niño extending into early 2027 is close to certainty, raising the risk of extreme weather events and supply disruptions. Given that India imports over 55 per cent of its edible oil requirements, elevated global prices and currency weakness continue to keep domestic edible oil inflation elevated at around 20 per cent. The monsoon impact on agricultural output and food prices will remain one of the most important variables to monitor this year.

> Economic history suggests that when supply-driven price pressures coincide with robust growth, inflation often broadens beyond a narrow set of categories

Inflation risks, however, are no longer confined to supply-side shocks. Demand-side pressures also appear stronger than previously anticipated. GDP growth in the first quarter was not only above trend but also exceeded previous expectations, and the second quarter is also doing well. Reflecting this momentum, the RBI has raised its growth forecast for this fiscal to 7.1 per cent from 6.7 per cent earlier. Crisil likewise expects growth of 7.0 per cent. The interaction between persistent supply shocks and strong demand warrants particular attention. Economic history suggests that when supply-driven price pressures coincide with robust growth, inflation often broadens beyond a narrow set of categories. Early signs of such generalisation are becoming visible in core inflation, with price pressures gradually spreading beyond food and fuel.

Global developments reinforce the need for caution. Bond yields in advanced economies have risen much more sharply than in India, reflecting persistent inflation and growing concerns over fiscal sustainability. Consequently, the spread between US and Indian 10-year government bond yields has narrowed to roughly 200 basis points (bps), compared with over 400 a few years ago, reducing the relative attractiveness of Indian government securities for foreign investors. At the same time, major central banks remain in tightening mode. The US Federal Reserve raised rates by 25 bps in September, and markets continue to price in further tightening. The European Central Bank and the Bank of Japan have also maintained a tightening bias in response to persistent inflation pressures.

India's liquidity situation adds another layer of complexity. The RBI's efforts to mobilise foreign currency non-resident deposits have been highly successful, strengthening foreign exchange reserves and improving balance-of-payments resilience. But these inflows have also injected substantial liquidity into the financial system. Consequently, financial conditions eased through August, and liquidity remains abundant. This creates a degree of tension, albeit transitory, between prevailing monetary conditions and the tightening bias increasingly warranted by growth and inflation dynamics.

Taken together, these developments explain the RBI's policy shift. Inflation risks are becoming more broad-based, growth remains resilient, global monetary conditions continue to tighten, and domestic liquidity remains ample. Policymakers will need to remain highly vigilant on price developments. While future actions will remain data-dependent, the possibility of another rate hike this year can no longer be ruled out.

The writer is chief economist, Crisil

Where light stops, neutrinos begin



BIMAN NATH

FOR AN astronomer, light is the easy part: Hold up a lens or a mirror and starlight is yours. Neutrinos are another matter. Born in the most violent places in the cosmos, these ghost-like particles pass through most detectors almost without a trace. It took decades to turn a cubic kilometre of Antarctic ice into a telescope for these deep-space messengers. This year's Nobel Prize in Physics goes to the man who stayed with that hunt: Francis Halzen, a Belgian-born physicist at the University of Wisconsin-Madison.

Why chase a particle so hard to catch? "You could see things in the universe you couldn't see any other way," Halzen said. Neutrinos barely interact, so they escape from deep inside their sources while light gets stuck on the way out. Think of a journalist trying to reach the one official who was in the room where a major policy was made. The press conference is easy to attend; that is astronomy with visible light. But the insider who knows what was really decided avoids reporters. Catching a neutrino is harder still.

The IceCube Neutrino Observatory in Antarctica has lowered 86 cables into the ice, the deepest reaching almost 2.5 km down. Each cable sports 60 sensitive light detectors, like beads strung on a thread. On rare occasions, a neutrino strikes a nucleus of an atom in the ice, and the collision generates high-energy particles. The particles move faster than light travels in ice, and generate a bluish flash, like a sonic boom of a supersonic jet. That glow, the signature of a far-travelling neutrino, is what the sensors pick up.

Such an instrument couldn't be built overnight. What Halzen himself once called a "cute idea" took smaller trials to hone the skill and show what a larger detector could potentially deliver. Conceived in the 1980s, the last cable in IceCube was dropped in 2010. Since then, it has been eavesdropping on far-away violent phenomena in the universe. Some strange galaxies, unlike our Milky Way, host massive black holes that churn the gas in their cores and spew spectacular jets. These accelerate high-energy particles called cosmic rays, which collide and produce neutrinos. These neutrinos point back to where cosmic rays are produced.

The discoveries at IceCube have been helping astronomers peek inside the core of these galaxies. In 2017, a single neutrino hurtled through the observatory. Its path indicated it was coming from the direction of the constellation Orion. Astronomers swiveled their telescopes in that direction and found a "blazar", a giant galaxy whose central black hole, weighing hundreds of millions of suns, spits out a jet aimed at Earth. The discovery gave a major boost to "multi-messenger" astronomy, where scientists use visible light, neutrinos, and gravitational waves to study the same cosmic event, pulling together the clues from "all the light we cannot see". Halzen, the principal investigator of this venture, has long been drawn to astroparticle physics, the area of research that marries astronomy with particle physics. But IceCube was never a one-man project: It is the work of an international collaboration of hundreds of scientists and engineers. Starlight can be caught with a mirror. Neutrinos take patience. IceCube received its first major expansion this year, and physicists are proposing a successor eight times its size. Somewhere out there, another neutrino is on its way. Thanks to Halzen and team, the ice is ready.

> Why chase a particle so hard to catch? 'You could see things in the universe you couldn't see any other way,' Halzen said

The writer, visiting professor, IISER Mohali, was an astrophysicist at Raman Research Institute, Bengaluru